

$$F(A, B, C, D) = \Sigma m(1, 2, 9, 11, 15) + \Sigma d(4, 13)$$

NEW FUNCTION

2. Finding Prime Implicants

Group	Decimal	Binary
0	1	0001 ✓
0	2	0010 ✖
0	4	0100 ✖
1	9	1001 ✓
2	11	1011 ✓
2	13	1101 ✓
3	15	1111 ✓

3. Finding Prime Implicants

Group	Decimal	Binary
0	1, 9	-001 ✖
1	13, 9	1-01 ✓

Group	Decimal	Binary
1	11, 9	10-1 ✓
2	11, 15	1-11 ✓
2	13, 15	11-1 ✓

4. Finding Prime Implicants

Group	Decimal	Binary
0	11, 13, 15, 9	1--1 ✱

The Prime Implicants Are
A'B'CD', A'BC'D', B'C'D, AD

6. Prime Implicants Chart

Minterms	1	11	15	2	9	Cost
A'B'CD'				X		8
B'C'D	X				X	6
AD		X	X		X	3

7. Finding Unique Minterms

Minterms	1	11	15	2	9	Cost
$A'B'CD'$				X		8
$B'C'D$	X				X	6
AD		X	X		X	3

The Essential Implicants Are
 $A'B'CD'$, $B'C'D$, AD

Possible Function Minimizations
 $F = A'B'CD' + B'C'D + AD$